

Easy Level MCQs:**

Easy Level MCQs:

1. What does the `chmod` command do in Linux?

- a) Change file owner
- b) Change file permissions
- c) Change file group
- d) Change file timestamp

Answer: b

2. Which command is used to connect to a remote server using SSH?

- a) telnet
- b) ftp
- c) ssh
- d) scp

Answer: c

3. How can you list all users on a Linux system?

- a) `cat /etc/passwd`
- b) `ls -l /home`
- c) `ps aux`
- d) `who`

Answer: a

4. What is the primary purpose of the Kickstart tool in Linux?

- a) Automate Linux installation
- b) Manage software packages
- c) Monitor system performance
- d) Troubleshoot network issues

Answer: a

5. What does the `chown` command do?

- a) Changes file permissions
- b) Changes file ownership
- c) Changes file timestamp
- d) Changes file extension

Answer: b

Medium Level MCQs:

1. What is the correct syntax to grant execute permission to all users for a file?

- a) `chmod +x filename`
- b) `chmod a+x filename`

c) `chmod ugo+x filename`

d) `chmod 777 filename`

Answer: b

2. Which file stores information about groups in Linux?

a) `/etc/passwd`

b) `/etc/group`

c) `/etc/shadow`

d) `/etc/users`

Answer: b

3. How do you restrict access to a file using ACL?

a) `setfacl -m u:user:r file`

b) `chmod 755 file`

c) `chown user file`

d) `usermod -aG file user`

Answer: a

4. Which network command is used for secure file transfer?

a) ftp

b) sftp

c) telnet

d) rsync

Answer: b

5. What does the Kickstart `url` directive specify?

a) URL for package repository

b) URL for the Kickstart configuration file

c) URL for downloading system updates

d) URL for system error logs

Answer: b

Difficult Level MCQs:

1. How can you recursively modify permissions for directories and files separately?

a) `chmod -R 755 *`

b) `find . -type d -exec chmod 755 {} \;`

c) `chown -R user *`

d) `chmod --recursive 755 *`

Answer: b

2. What is the difference between `passwd` and `chage` commands?

a) `passwd` changes passwords while `chage` manages password expiration

b) Both change passwords

c) `passwd` manages password expiration and `chage` locks accounts

d) Both manage expiration

Answer: a

3. **How do you allow multiple users to have write access to a directory without altering other permissions?**

a) Use setfacl

b) Use chmod

c) Use chown

d) Use umask

Answer: a

4. **What is the purpose of the `kickstart --useexisting` option?**

a) Use an existing installation configuration

b) Restore a previous system state

c) Reboot after installation

d) Reuse the current package manager settings

Answer: a

5. **How do you test SSH key authentication without entering a password?**

a) `ssh-keygen -t rsa`

b) `ssh-copy-id user@host`

c) `scp -i keyfile user@host`

d) `telnet user@host`

Answer: b

Easy Level MCQs:

6. **Which tool is commonly used for graphical Linux installation?**

a) YUM

b) Anaconda

c) RPM

d) Kickstart

Answer: b

7. **What is the purpose of the GRUB boot loader?**

a) Manage user accounts

b) Load the operating system kernel into memory

c) Manage network configurations

d) Manage system packages

Answer: b

8. **What file is commonly used to configure GRUB boot settings?**

a) `/etc/fstab`

b) `/boot/grub/grub.cfg`

c) `/etc/grub.conf`

d) `/boot/initramfs`

Answer: b

9. **What is the purpose of the Initial RAM Disk (initramfs)?**

- a) To permanently store the operating system
- b) To initialize the kernel modules needed to boot the system
- c) To store user home directories
- d) To configure network settings during boot

Answer: b

10. **Which run level is used for multi-user mode with networking enabled?**

- a) Run Level 0
- b) Run Level 1
- c) Run Level 3
- d) Run Level 6

Answer: c

Medium Level MCQs:

1. **Which of the following is a method for performing a hands-free Linux installation?**

- a) Anaconda Text Mode
- b) Kickstart
- c) GRUB Boot Menu
- d) YUM

Answer: b

2. **Which command can be used to rebuild the GRUB configuration file on a modern Linux system?**

- a) `grub-install`
- b) `update-grub`
- c) `mkinitrd`
- d) `dracut`

Answer: b

3. **What does the run level 5 represent in traditional Linux systems?**

- a) Reboot
- b) Single-user mode
- c) Multi-user mode with GUI
- d) Halt the system

Answer: c

4. **Which of these tools can be used to create an Initial RAM Disk (initramfs)?**

- a) `grub2-mkconfig`
- b) `dracut`
- c) `grub-install`
- d) `rpm`

Answer: b

5. Where is the GRUB boot loader commonly installed on a BIOS-based system?

- a) `/boot` partition
- b) Master Boot Record (MBR)
- c) `/etc/grub.conf`
- d) `/var/lib`

Answer: b

Difficult Level MCQs:

1. Which directive in the Kickstart file defines the disk partitioning layout?

- a) `%packages`
- b) `%pre`
- c) `part`
- d) `bootloader`

Answer: c

2. How do you change the default kernel boot entry in GRUB 2?

- a) Modify `/etc/default/grub` and run `update-grub`
- b) Edit `/boot/grub/grub.cfg` directly
- c) Use the `grub-install` command
- d) Modify `/etc/fstab`

Answer: a

3. What is the purpose of the `init` process during the Linux boot sequence?

- a) Loads the kernel into memory
- b) Mounts the root filesystem
- c) Initializes system processes and services
- d) Generates the `initramfs` file

Answer: c

4. Which command can you use to view the current run level on a Linux system using `systemd`?

- a) `runlevel`
- b) `systemctl get-default`
- c) `who -r`
- d) `initctl list`

Answer: b

5. What happens if the GRUB bootloader is corrupted on a BIOS-based system?

- a) The system boots into rescue mode
- b) The system will fail to boot and show a GRUB error
- c) The kernel will load but services won't start
- d) The system will boot into single-user mode

Answer: b

Easy Level MCQs:

6. Which command is used to restart a service in a system using `systemd`?

- a) `service restart <service_name>`
- b) `systemctl restart <service_name>`
- c) `restart <service_name>`
- d) `init.d <service_name> restart`

Answer: b

7. What is the primary purpose of OpenSSH?

- a) Remote desktop sharing
- b) Secure remote shell access
- c) Web server management
- d) System performance monitoring

Answer: b

8. Which file contains system-wide environment variables in most Linux distributions?

- a) `/etc/fstab`
- b) `/etc/passwd`
- c) `/etc/profile`
- d) `/etc/hostname`

Answer: c

9. What is the default port used by the VNC server?

- a) 21
- b) 22
- c) 5900
- d) 8080

Answer: c

10. What does the X Window System (X11) primarily provide in a Linux system?

- a) Command-line interface
- b) Graphical user interface
- c) Network firewall management
- d) Software package installation

Answer: b

Medium Level MCQs:

1. Which command is used to display the current status of all services using `systemd`?

- a) `service --status-all`
- b) `systemctl list-units --type=service`
- c) `chkconfig --list`
- d) `statusctl all`

Answer: b

2. Which configuration file is modified to define SSH access restrictions?

- a) `/etc/ssh/sshd_config`

b) `/etc/hosts.allow`

c) `/etc/passwd`

d) `/etc/ssh_config`

Answer: a

3. Which tool can be used to manage the firewall for allowing VNC connections on a Linux system?

a) `iptables`

b) `firewalld`

c) `ufw`

d) All of the above

Answer: d

4. What is the role of the `/etc/x11/xorg.conf` file?

a) Configures kernel boot options

b) Manages system services

c) Configures the graphical display settings

d) Controls network authentication

Answer: c

5. Which option can disable root login over SSH in the `sshd_config` file?

a) `PermitRootLogin no`

b) `AllowRootLogin no`

c) `RootLogin no`

d) `DenyRootAccess no`

Answer: a

Difficult Level MCQs:

1. How can you create a custom service unit file for `systemd`?

a) Modify `/etc/systemd.conf`

b) Create a `.service` file in `/etc/systemd/system/`

c) Use `systemctl create service`

d) Edit `/etc/init.d/`

Answer: b

2. What does the `Match` directive do in an `sshd_config` file?

a) Enables SSH key authentication

b) Applies conditional settings for specific users, groups, or networks

c) Matches allowed IP ranges

d) Matches specific SSH keys

Answer: b

3. How can you test the X server configuration without restarting the system?

a) `startx --test`

b) `Xorg -configure`

c) `X -retro`

d) `xinit`

Answer: b

4. Which protocol does VNC use for remote desktop sharing?

a) RDP

b) VNC

c) RFB

d) X11

Answer: c

5. What is the purpose of the `systemctl daemon-reexec` command?

a) Restart the entire system

b) Re-execute the systemd daemon without rebooting

c) Reset all service states

d) Flush all service logs

Answer: b

Easy Level MCQs:

6. Which command is used to create a new partition on a disk in Linux?

a) `mkfs`

b) `fdisk`

c) `mount`

d) `lvm`

Answer: b

7. What is the primary purpose of the `LVM` (Logical Volume Manager)?

a) Create a bootloader

b) Manage logical volumes for flexible storage

c) Manage user accounts

d) Format network drives

Answer: b

8. Which of the following commands can be used to format a partition with the ext4 file system?

a) `fdisk -t ext4`

b) `mkfs.ext4`

c) `lvm format`

d) `mount -t ext4`

Answer: b

9. Which service is commonly used for print service management in Linux?

a) `CUPS`

b) `LVM`

c) `fdisk`

d) `NFS`

Answer: a

10. What is the default device name for the first hard disk in Linux?

a) `/dev/sda`

b) `/dev/hda`

c) `/dev/disk1`

d) `/mnt/hda`

Answer: a

Medium Level MCQs:

1. Which command is used to display information about existing partitions on a disk?

a) `lsblk`

b) `df`

c) `gdisk`

d) `blkid`

Answer: a

2. What does the `vgcreate` command do in the context of LVM?

a) Create a physical volume

b) Create a logical volume

c) Create a volume group

d) Create a file system

Answer: c

3. Which file must be updated after creating a new partition for it to mount automatically on boot?

a) `/etc/fstab`

b) `/boot/grub.cfg`

c) `/etc/partitions`

d) `/proc/mounts`

Answer: a

4. How can you check the status of a print job in a CUPS-managed printer?

a) `lpstat -p`

b) `cups list`

c) `service cups status`

d) `printstat`

Answer: a

5. Which of the following tools can be used for GUID partition table (GPT) management?

a) `fdisk`

b) `gdisk`

c) `parted`

d) Both b and c

Answer: d

Difficult Level MCQs:

1. What is the difference between `fdisk` and `gdisk`?

- a) `fdisk` supports MBR, while `gdisk` supports GPT
- b) `fdisk` supports both MBR and GPT
- c) `gdisk` is only for removable drives
- d) Both manage RAID arrays

Answer: a

2. Which command can be used to extend an existing logical volume in LVM?

- a) `lvcreate`
- b) `lvextend`
- c) `vgextend`
- d) `lvresize`

Answer: b

3. Which of the following steps is required to format and mount a secondary storage device?

- a) Create a partition, format it, and mount the partition
- b) Format the disk and reboot the system
- c) Create a partition and modify `/etc/passwd`
- d) Install a bootloader and run `fdisk`

Answer: a

4. How can you share a printer over a network using CUPS?

- a) Modify `/etc/printcap` and restart the service
- b) Enable printer sharing in the CUPS web interface or `cupsctl --share`
- c) Use `lpd` for network printing
- d) Modify `/etc/cups/cupsd.conf` and restart the service

Answer: b

5. What does the `pvcreate` command do in an LVM setup?

- a) Create a physical volume for use in a volume group
- b) Create a partition on a physical disk
- c) Create a virtual disk for testing
- d) Manage print queues

Answer: a

Easy Level MCQs:

6. Which directory contains most of the system configuration files in a Linux system?

- a) `/home`
- b) `/etc`
- c) `/var`

d) `/usr`

Answer: b

7. What is the purpose of a patch in system maintenance?

- a) To clean disk space
- b) To update or fix software vulnerabilities
- c) To create a new user
- d) To format the system

Answer: b

8. Which command can be used to update all installed packages on a system using `yum`?

- a) `yum fetch`
- b) `yum update`
- c) `yum upgrade`
- d) `yum install`

Answer: b

9. What file should be edited to configure system-wide environment variables in Linux?

- a) `/etc/hosts`
- b) `/etc/profile`
- c) `/var/log/messages`
- d) `/boot/grub.cfg`

Answer: b

10. Which of the following is a patch management tool used in Red Hat-based systems?

- a) `apt-get`
- b) `yum`
- c) `zypper`
- d) `pacman`

Answer: b

Medium Level MCQs:

1. What is the purpose of the `/etc/fstab` file?

- a) Manage user accounts
- b) Define disk partitions and mount points
- c) List installed packages
- d) Control the GRUB bootloader

Answer: b

2. Which command can be used to list the currently loaded kernel patches?

- a) `uname -a`
- b) `dmesg`
- c) `rpm -qa | grep kernel`

d) `lsmod`

Answer: c

3. Which option in the `apt-get` command can be used to install security updates only?

a) `apt-get upgrade`

b) `apt-get dist-upgrade`

c) `apt-get install --only-upgrade`

d) `apt-get update --security`

Answer: c

4. What is the primary difference between the `/etc/passwd` and `/etc/shadow` files?

a) `/etc/passwd` stores passwords while `/etc/shadow` stores usernames

b) `/etc/passwd` stores usernames, while `/etc/shadow` stores encrypted passwords

c) `/etc/shadow` controls services and `/etc/passwd` manages patches

d) Both files store kernel patches

Answer: b

5. What is the purpose of the `yum clean all` command?

a) Remove all installed packages

b) Clear the package cache

c) Reboot the system

d) Delete unused kernel patches

Answer: b

Difficult Level MCQs:

1. Which command can be used to manually apply a patch file to a source code directory?

a) `patch -p1 < file.patch`

b) `diff -u file1 file2`

c) `git apply file.patch`

d) `rpm -ivh file.rpm`

Answer: a

2. How can you prevent a package from being updated during a system-wide update using `yum`?

a) `yum hold <package>`

b) `yum exclude <package>`

c) Add `exclude=<package>` to `/etc/yum.conf`

d) Both b and c

Answer: d

3. Which configuration file determines the runlevel or default system target in a `systemd`-based Linux system?

a) `/etc/inittab`

b) `/etc/systemd/system/default.target`

c) `/boot/grub.cfg`

d) `/etc/rc.local`

Answer: b

4. When using `rpm`, how can you verify if a specific patch has already been installed?

a) `rpm -qa | grep <package>`

b) `rpm -q --changelog <package>`

c) `yum list installed <package>`

d) `rpm --verify <package>`

Answer: b

5. What does the `/etc/sysctl.conf` file control in Linux?

a) Kernel parameters at runtime

b) Network configurations

c) Patch versioning

d) GRUB configurations

Answer: a

Easy Level (10 Questions)

1. Which command is used to check the syntax of a DNS zone file in BIND?

a) `named-checkconf`

b) `named-checkzone`

c) `dig`

d) `nslookup`

Answer: b

2. Which file is commonly edited for DNS zone configurations in a BIND server?

a) `/etc/hosts`

b) `/etc/named.conf`

c) `/etc/resolv.conf`

d) `/etc/zone.conf`

Answer: b

3. What port does the FTP service use by default?

a) 21

b) 22

c) 23

d) 25

Answer: a

4. Which protocol is used for secure remote access in Linux systems?

a) FTP

b) SSH

c) Telnet

d) HTTP

Answer: b

5. What is the primary purpose of the `postfix` service?

- a) File sharing
- b) Email transfer
- c) Web hosting
- d) Network monitoring

Answer: b

6. Which configuration file is used to set up a basic Apache web server?

- a) `/etc/httpd.conf`
- b) `/etc/apache2/apache2.conf`
- c) `/etc/httpd/httpd.conf`
- d) `/var/www/index.html`

Answer: b

7. What is RAID 1 commonly known for?

- a) Striping
- b) Mirroring
- c) Parity
- d) Deduplication

Answer: b

8. Which service is used to share files between Linux and Windows systems?

- a) NFS
- b) Samba
- c) FTP
- d) SCP

Answer: b

9. What does the `dovecot` service provide?

- a) Web hosting
- b) Database management
- c) IMAP and POP3 mail services
- d) Firewall protection

Answer: c

10. Which command can restart the Apache web server?

- a) `systemctl restart apache2`
- b) `service restart httpd`
- c) `apachectl restart`
- d) All of the above

Answer: d

1. Which file defines the primary and secondary name servers in a BIND DNS setup?

- a) `/etc/resolv.conf`
- b) `/etc/named.rfc1912.zones`
- c) `/etc/named.conf`
- d) `/etc/hosts`

Answer: c

2. What is the difference between NFSv3 and NFSv4?

- a) NFSv4 supports stateful connections, while NFSv3 does not
- b) NFSv4 only supports Windows clients
- c) NFSv3 supports encryption, while NFSv4 does not
- d) NFSv4 requires additional software for file sharing

Answer: a

3. Which directive in the Apache configuration file enables virtual hosting?

- a) `<VirtualHost>`
- b) `<Directory>`
- c) `Listen`
- d) `ServerRoot`

Answer: a

4. What is the primary difference between RAID 1 and RAID 5?

- a) RAID 1 offers mirroring while RAID 5 offers striping with parity
- b) RAID 5 mirrors data, while RAID 1 uses striping
- c) RAID 1 offers higher speed than RAID 5
- d) RAID 5 requires only two disks, while RAID 1 requires three

Answer: a

5. Which command would you use to test if a DNS server is responding correctly?

- a) `ping`
- b) `dig`
- c) `netstat`
- d) `traceroute`

Answer: b

6. How do you enable an Apache virtual host configuration in Ubuntu?

- a) `a2ensite <site-name>`
- b) `apachectl addhost`
- c) `systemctl addhost`
- d) `httpd -v`

Answer: a

7. Which tool is used to monitor and manage RAID arrays in Linux?

- a) `mdadm`
- b) `raidctl`
- c) `lvm`

d) parted

Answer: a

8. What is the default port used by the SSH service?

a) 20

b) 21

c) 22

d) 25

Answer: c

9. Which directive in `/etc/samba/smb.conf` defines a shared folder?

a) [folder]

b) [shared]

c) [homes]

d) [share]

Answer: d

10. What is the primary purpose of the DHCP server?

a) Assign IP addresses automatically to devices

b) Provide secure email transfer

c) Share files between systems

d) Manage firewall rules

Answer: a

Difficult Level (10 Questions)

1. How can you prevent DNS zone transfer for security reasons in BIND?

a) Add `zone_transfer none` in `named.conf`

b) Use `allow-transfer { none; };` in the zone file

c) Enable IP blocking using `iptables`

d) Remove the zone file

Answer: b

2. Which RAID level provides both striping and parity?

a) RAID 1

b) RAID 0

c) RAID 5

d) RAID 10

Answer: c

3. In Postfix, which file defines the allowed relay domains?

a) `/etc/postfix/main.cf`

b) `/etc/postfix/relay_domains`

c) `/etc/postfix/access`

d) `/etc/postfix/master.cf`

Answer: a

4. What is the purpose of the `chroot` jail in a DNS server?

- a) Secure the DNS server by isolating it from the system
- b) Run multiple DNS servers
- c) Disable recursive queries
- d) Block unauthorized users

Answer: a

5. Which command creates a RAID 5 array with 3 disks?

- a) `mdadm --create --level=5 --raid-devices=3 /dev/md0 /dev/sd{b,c,d}`
- b) `raid5 --init /dev/md0`
- c) `lvm create raid5`
- d) `parted --raid5`

Answer: a

6. Which configuration file in Apache specifies SSL/TLS settings?

- a) `/etc/apache2/ssl.conf`
- b) `/etc/apache2/sites-available/default-ssl.conf`
- c) `/etc/httpd/ssl.conf`
- d) `/etc/ssl/ssl.conf`

Answer: b

7. What does the `dovecot` service do for Postfix?

- a) Provides IMAP and POP3 services for mail delivery
- b) Filters spam emails
- c) Handles DNS lookups
- d) Encrypts web traffic

Answer: a

8. How do you assign a static IP address using DHCP?

- a) Configure a reservation with the MAC address in `dhcpd.conf`
- b) Use `ipconfig`
- c) Add a static route in `/etc/network/interfaces`
- d) Assign a reserved IP using Apache

Answer: a

9. Which command tests if an NFS share is accessible?

- a) `mount`
- b) `showmount`
- c) `exportfs`
- d) `nfsstat`

Answer: b

10. Which protocol is primarily used for directory services and network authentication?

- a) LDAP
- b) IMAP
- c) SMTP

d) POP3

Answer: a

Easy Level (5 Questions)

1. Which protocol is commonly used for sending emails?

- a) IMAP
- b) SMTP
- c) POP3
- d) FTP

Answer: b

2. What is the primary role of the Postfix mail server?

- a) Web hosting
- b) Database management
- c) Sending and receiving emails
- d) Managing DNS records

Answer: c

3. Which of the following ports is commonly used by IMAP?

- a) 110
- b) 143
- c) 80
- d) 22

Answer: b

4. What is the default port for the POP3 protocol?

- a) 25
- b) 110
- c) 587
- d) 995

Answer: b

5. What is the purpose of SquirrelMail?

- a) Sending bulk emails
- b) Web-based email client
- c) Spam filtering
- d) DNS management

Answer: b

Medium Level (5 Questions)

6. Which file is the primary configuration file for Postfix?

- a) `/etc/postfix/main.cf`
- b) `/etc/postfix/relay.conf`
- c) `/etc/postfix/smtpd.conf`

d) `/etc/postfix/mail.conf`

Answer: a

7. What does the `dovecot` service primarily provide?

- a) Web-based email access
- b) POP3 and IMAP services
- c) DNS resolution
- d) File sharing services

Answer: b

8. Which command can be used to restart the Postfix service in a Linux system?

- a) `systemctl restart postfix`
- b) `postfix restart`
- c) `service postfix start`
- d) `systemctl reload postfix`

Answer: a

9. What is the purpose of the `maildir` format in Postfix?

- a) It stores emails in a single file.
- b) It stores each email in a separate file.
- c) It compresses all emails into a ZIP file.
- d) It allows only outgoing emails.

Answer: b

10. Which Dovecot configuration file specifies authentication settings?

- a) `/etc/dovecot/dovecot.conf`
- b) `/etc/dovecot/auth.conf`
- c) `/etc/dovecot/dovecot.auth`
- d) `/etc/dovecot/imap.conf`

Answer: a

Difficult Level (5 Questions)

11. How can you restrict Postfix to send emails only from authenticated users?

- a) `smtpd_sender_restrictions = reject_unauth_destination`
- b) `smtp_sender_restrictions = allow_all`
- c) `smtpd_sender_restrictions = permit_all`
- d) `smtp_sender_restrictions = reject_unknown`

Answer: a

12. Which configuration file is edited to set virtual mailboxes in Postfix?

- a) `/etc/postfix/main.cf`
- b) `/etc/postfix/virtual`
- c) `/etc/postfix/master.cf`
- d) `/etc/postfix/mail.cf`

Answer: b

13. What is the purpose of the `master.cf` file in Postfix?

- a) Manage mail queues
- b) Define the services and daemons used by Postfix
- c) Configure DNS records
- d) Manage web hosting

Answer: b

14. Which of the following commands rebuilds the Postfix alias database?

- a) `postfix update`
- b) `newaliases`
- c) `postfix reload`
- d) `postfix upgrade`

Answer: b

15. What is the difference between IMAP and POP3 in Dovecot?

- a) IMAP downloads emails, while POP3 keeps them on the server.
- b) IMAP keeps emails on the server, while POP3 downloads them.
- c) IMAP is faster than POP3.
- d) POP3 supports multiple devices, while IMAP does not.

Answer: b

Easy Level MCQs

1. Which command is used to monitor CPU usage in Linux?

- a) top
- b) ps
- c) cp
- d) grep

Answer: a) top

2. What does NTP stand for in Linux?

- a) Network Time Processor
- b) Network Time Protocol
- c) Network Troubleshooting Protocol
- d) Network Timing Processor

Answer: b) Network Time Protocol

3. Which file typically stores system logs in Linux?

- a) `/etc/logs/system`
- b) `/var/log/syslog`
- c) `/usr/log/messages`
- d) `/tmp/system.log`

Answer: b) `/var/log/syslog`

4. What is the primary purpose of an audit log?

- a) To store user data
- b) To monitor and record security-related events

- c) To update system software
- d) To configure hardware settings

Answer: b) To monitor and record security-related events

5. **Which command can synchronize the system clock with an NTP server?**

- a) ntpdate
- b) timedatectl
- c) clockctl
- d) timefix

Answer: a) ntpdate

Medium Level MCQs

6. **Which command is used to analyze disk I/O performance?**

- a) iostat
- b) dstat
- c) diskperf
- d) ioanalyze

Answer: a) iostat

7. **What is the default port used by the NTP protocol?**

- a) 23
- b) 68
- c) 123
- d) 143

Answer: c) 123

8. **Which of the following is used for logging kernel messages in Linux?**

- a) syslog
- b) journald
- c) dmesg
- d) rsyslog

Answer: c) dmesg

9. **What is the purpose of configuring "noexec" on a partition?**

- a) To allow scripts to execute
- b) To prevent the execution of binaries
- c) To enhance network communication
- d) To increase disk read speed

Answer: b) To prevent the execution of binaries

10. **Which file is primarily used to configure the NTP server?**

- a) /etc/ntp.conf
- b) /var/lib/ntp/server.conf
- c) /usr/ntp/settings.conf
- d) /etc/systemd/ntp.conf

Answer: a) /etc/ntp.conf

Difficult Level MCQs

11. Which Linux tool allows real-time monitoring of performance metrics like CPU, memory, and network usage?

- a) htop
- b) sar
- c) iotop
- d) perf

Answer: b) sar

12. What is the purpose of the "relatime" mount option in Linux?

- a) To disable access time updates completely
- b) To update access times only if they are older than the modification time
- c) To enable real-time performance tracking
- d) To allow read and write access simultaneously

Answer: b) To update access times only if they are older than the modification time

13. How does setting up an NTP stratum level help in time synchronization?

- a) It defines the maximum number of peers allowed.
- b) It indicates the distance from the reference clock.
- c) It determines the number of retries for synchronization.
- d) It restricts the use of local clocks.

Answer: b) It indicates the distance from the reference clock.

14. What does the "auditd" daemon do in Linux?

- a) Manages package updates
- b) Collects and logs audit data related to security events
- c) Provides time synchronization services
- d) Optimizes disk usage automatically

Answer: b) Collects and logs audit data related to security events

15. Which configuration file is used for customizing systemd journal log settings?

- a) /etc/systemd/journald.conf
- b) /etc/journal/log.conf
- c) /var/log/systemd.conf
- d) /usr/lib/systemd/journal.conf

Answer: a) /etc/systemd/journald.conf

Easy Level MCQs

1. What does BIND stand for in the context of DNS?

- a) Basic Internet Node Directory
- b) Berkeley Internet Name Domain
- c) Binary Internet Name Database
- d) Basic Information Name Domain

Answer: b) Berkeley Internet Name Domain

2. Which port does DNS use for queries?

- a) 25
- b) 53
- c) 110
- d) 143

Answer: b) 53

3. What is the role of an LDAP server?

- a) To handle DNS queries
- b) To authenticate and store directory-based data
- c) To manage load balancing
- d) To provide email services

Answer: b) To authenticate and store directory-based data

4. What does the 'balancer' module in Apache do?

- a) Monitors server disk usage
- b) Distributes traffic among backend servers
- c) Secures files during transfer
- d) Compresses HTTP responses

Answer: b) Distributes traffic among backend servers

5. Which protocol does NIS primarily use for communication?

- a) TCP/IP
- b) RPC
- c) HTTP
- d) SMTP

Answer: b) RPC

Medium Level MCQs

6. Which file contains the primary configuration for a BIND DNS server?

- a) /etc/bind/config.dns
- b) /etc/named.conf
- c) /var/named/dns.conf
- d) /usr/local/bind/bind.conf

Answer: b) /etc/named.conf

7. What is a split-horizon DNS configuration?

- a) A DNS setup where different responses are given based on the querying user's location.
- b) A DNS server configured with multiple IP addresses.
- c) A DNS system that splits traffic between internal and external users.
- d) A DNS redundancy strategy involving multiple servers.

Answer: a) A DNS setup where different responses are given based on the querying user's location.

8. Which authentication mechanism does LDAP support for secure communication?

- a) Plaintext authentication
- b) Kerberos authentication

- c) OAuth
- d) OpenID

Answer: b) Kerberos authentication

9. In an Apache cluster, what is the primary role of the load balancer?

- a) To store user session data
- b) To distribute incoming traffic evenly across servers
- c) To serve static content only
- d) To secure connections using SSL

Answer: b) To distribute incoming traffic evenly across servers

10. What is the primary advantage of DNSSEC in DNS security?

- a) It encrypts DNS queries.
- b) It validates the authenticity of DNS responses.
- c) It hides DNS server IP addresses.
- d) It restricts DNS to internal networks only.

Answer: b) It validates the authenticity of DNS responses.

Difficult Level MCQs

11. Which DNSSEC record type is used to store a digital signature?

- a) DNSKEY
- b) RRSIG
- c) DS
- d) NSEC

Answer: b) RRSIG

12. How does the "round-robin" DNS load balancing technique work?

- a) It routes traffic based on the client's geographic location.
- b) It alternates DNS responses among multiple A records.
- c) It uses the shortest network path to determine the server.
- d) It chooses the least loaded server dynamically.

Answer: b) It alternates DNS responses among multiple A records.

13. Which configuration file is typically used to secure NIS communications with a specific set of IPs?

- a) /etc/yp.conf
- b) /var/nis/security.conf
- c) /etc/security/ypsec.conf
- d) /etc/nis/hosts.allow

Answer: a) /etc/yp.conf

14. What is the purpose of the `mod_proxy` module in Apache clustering?

- a) To enable SSL communication
- b) To act as a reverse proxy for backend servers
- c) To cache HTTP responses

d) To compress HTTP requests

Answer: b) To act as a reverse proxy for backend servers

15. **Which of the following helps prevent zone transfer attacks in BIND DNS?**

a) Enabling recursive queries

b) Limiting zone transfers to specific IPs

c) Using plaintext zone files

d) Disabling DNSSEC

Answer: b) Limiting zone transfers to specific IPs

Easy Level MCQs

1. **What does the `#!/bin/bash` line at the top of a script indicate?**

a) It is a comment line with no purpose.

b) It specifies the script uses the Bash shell.

c) It disables debugging.

d) It sets the default variable scope.

Answer: b) It specifies the script uses the Bash shell.

2. **Which command is used to display output in a Bash script?**

a) print

b) show

c) echo

d) display

Answer: c) echo

3. **How do you assign a value to a variable in Bash?**

a) `$variable = "value"`

b) `variable == "value"`

c) `variable="value"`

d) `variable::"value"`

Answer: c) `variable="value"`

4. **Which operator is used for string comparison in Bash?**

a) `==`

b) `&&`

c) `||`

d) `//`

Answer: a) `==`

5. **What does the `-e` option in the `echo` command do?**

a) Enables special character interpretation.

b) Exits the script after displaying output.

c) Suppresses output to the terminal.

d) Displays errors on the screen.

Answer: a) Enables special character interpretation.

Medium Level MCQs

6. What does the `set -e` command do in a Bash script?

- a) Suppresses errors in the script.
- b) Exits the script immediately on error.
- c) Logs all errors to a file.
- d) Enables extended debugging.

Answer: b) Exits the script immediately on error.

7. Which Bash command is used to debug a script line-by-line?

- a) `bash -d`
- b) `bash -v`
- c) `bash -x`
- d) `bash --debug`

Answer: c) `bash -x`

8. What does the `[[` command provide in Bash scripting compared to `[`?

- a) Faster execution of conditional statements.
- b) Additional operators for advanced conditionals.
- c) Support for string concatenation.
- d) Logging of conditions during execution.

Answer: b) Additional operators for advanced conditionals.

9. What is the difference between `>` and `>>` in Bash?

- a) `>` overwrites a file; `>>` appends to it.
- b) `>` appends to a file; `>>` overwrites it.
- c) Both create a new file.
- d) `>>` redirects input; `>` redirects output.

Answer: a) `>` overwrites a file; `>>` appends to it.

10. Which loop is used to iterate over a list of items in Bash?

- a) `for`
- b) `while`
- c) `until`
- d) `do-while`

Answer: a) `for`

Difficult Level MCQs

11. How do you match a string that starts with "hello" and ends with "world" using a regular expression?

- a) `^hello world$`
- b) `^hello.*world$`
- c) `.*hello.world.`
- d) `^helloworld`

Answer: b) `^hello.*world$`

12. In Bash scripting, how can you handle an error in a command without stopping the script?

- a) Use `set +e` to ignore errors temporarily.
- b) Use `trap` to catch and handle the error.
- c) Use `set -e` to allow errors to propagate.
- d) Use `continue` to bypass the error.

Answer: a) Use `set +e` to ignore errors temporarily.

13. Which of the following correctly assigns and evaluates a variable within a conditional statement?

- a) `if [ var = "value" ]; then`
- b) `if [[ var == "value" ]]; then`
- c) `if ["var" == "value" ]; then`
- d) `if [ var = value ]; then`

Answer: c) `if [ "$var" == "value" ]; then`

14. How do you redirect both standard output and standard error to a single file in Bash?

- a) `command > file 2>&1`
- b) `command 2> file 1>&2`
- c) `command > file && 2>&1`
- d) `command > file |&`

Answer: a) `command > file 2>&1`

15. What does the command `trap "rm temp.txt" EXIT` do in a script?

- a) Deletes `temp.txt` when the script finishes or exits.
- b) Removes `temp.txt` immediately after execution.
- c) Deletes `temp.txt` only on errors.
- d) Removes `temp.txt` only on successful execution.

Answer: a) Deletes `temp.txt` when the script finishes or exits.
